Making a learned intent catalogue executable

Token-level prerequisite support, 21 July 2026

TextJEPA research cycle

The everyday problem: do I have the prerequisite?

- The controller sees a shuffled box of instruction cards called intent phrases.
- An instruction is executable only after its prerequisite computations have run.
- Compressing each earlier instruction into one vector can blur the exact entity names.
- We therefore let candidate words attend directly to words in the executed intent history.
- Success here means proposing executable actions without consulting a hidden feasible-action menu.

Token history repairs the proposal interface

Controller	Strict	Plus two	Invalid
Token head, history masked	.025	.025	.950
Phrase-pooled history	.083	.358	.483
Token-level history	**.175**	**.717**	**.158**

What to notice: token history roughly doubles the best observed phrase-pooled length-nine success and cuts invalid attempts by about two thirds. Why it matters: JEPA simulation can now be tested behind a viable, non-oracle proposal interface.

The scientific boundary is essential

- These are 120-episode, one-training-seed validation results.
- Every controller sees the same complete, shuffled, outcome-free intent catalogue.
- The negative control has equal token-head capacity but cannot see executed intent tokens.
- Selection was prior-only: JEPA consequence reranking had weight zero.
- The result supports lexical prerequisite tracking, not a claim that latent simulation improved planning.

Next decision: does simulation add value after support is fixed?

- Freeze the learning-rate-10^{-3} token-support checkpoint and the action catalogue.
- Compare prior-only selection with one-step and deeper causal JEPA reranking.
- Hold top-M, beam width, support weight, episodes, and problem order fixed.
- Scale only if reranking raises success without restoring invalid actions.
- If reranking hurts, separate value error from recursive world-model drift before retraining.